Deep Coherence Disentanglement (DCD): Single-shot Ultrasound Image Denoising by Pyramid Self-Contrastive Learning

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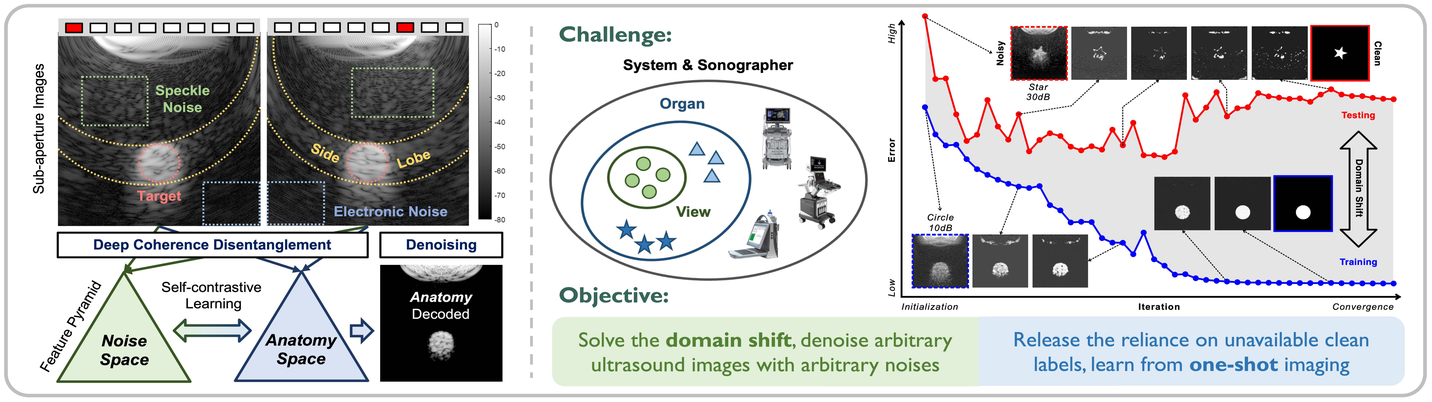
Reference

Contact

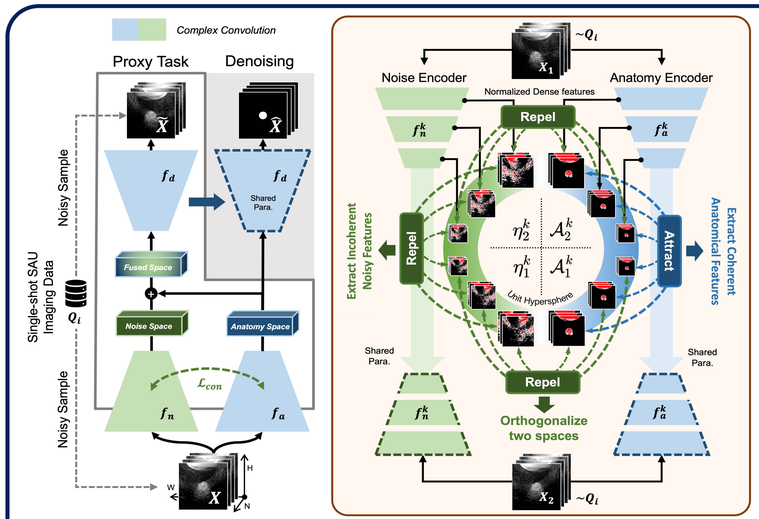
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❑ Overview



❑ Methodology



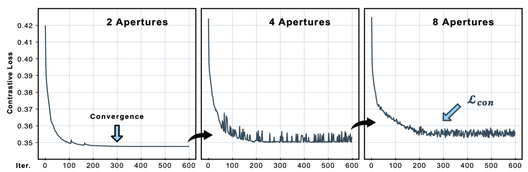
❖ **Cross-aperture coherence on raw IQ signal**

$$\mathcal{C}_{nm}(\boldsymbol{p}) = \left|\frac{\langle X_n X_m^* \rangle_{\boldsymbol{p}}}{\sqrt{\langle |X_n|^2 \rangle_{\boldsymbol{p}} \langle |X_m|^2 \rangle_{\boldsymbol{p}}}}\right|^2 = \mathrm{Cos}(\boldsymbol{x}_m, \boldsymbol{x}_n) = \frac{|\boldsymbol{x}_m^{\mathrm{H}} \cdot \boldsymbol{x}_n|^2}{\|\boldsymbol{x}_n\|_F^2 \cdot \|\boldsymbol{x}_m\|_F^2}, \ \boldsymbol{x}_{n/m} \subset X_{n/m}$$

❖ **Contrastive objectives in dense latent space**

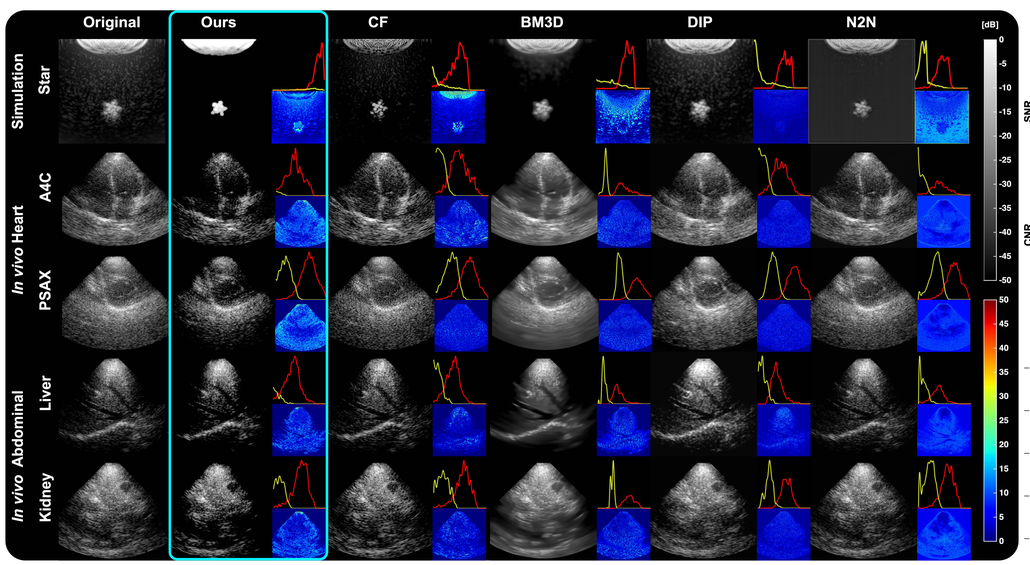
1. $\max_{f_a} \sum_{k=1}^{K} \sum_{Q_M} \mathrm{Cos}(\mathcal{A}_i^k, \mathcal{A}_j^k) = \sum_{k=1}^{K} \sum_{Q_M} \frac{|(\mathcal{A}_i^k)^{\mathrm{H}} \cdot \mathcal{A}_j^k|^2}{\|\mathcal{A}_i^k\|_F^2 \cdot \|\mathcal{A}_j^k\|_F^2}$ — *Attract coherent anatomical embeddings*

2. $\min_{f_n} \sum_{k=1}^{K} \sum_{Q_M} \mathrm{Cos}(\eta_i^k, \eta_j^k) = \sum_{k=1}^{K} \sum_{Q_M} \frac{|(\eta_i^k)^{\mathrm{H}} \cdot \eta_j^k|^2}{\|\eta_i^k\|_F^2 \cdot \|\eta_j^k\|_F^2}$ — *Repel incoherent noisy embeddings*

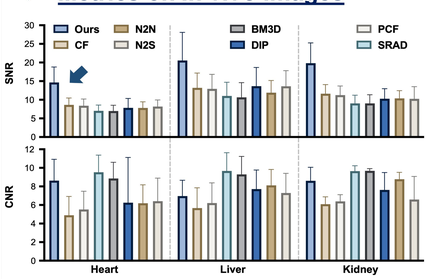


Increasing sub-aperture num. → *cross-aperture coherence ↓* → *Elevating convergence level*

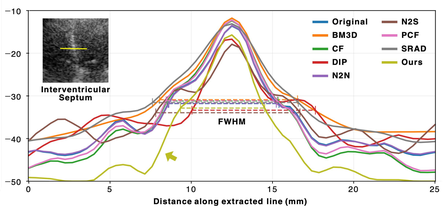
❑ Results Comparison



❖ **Metrics on *In vivo* images**

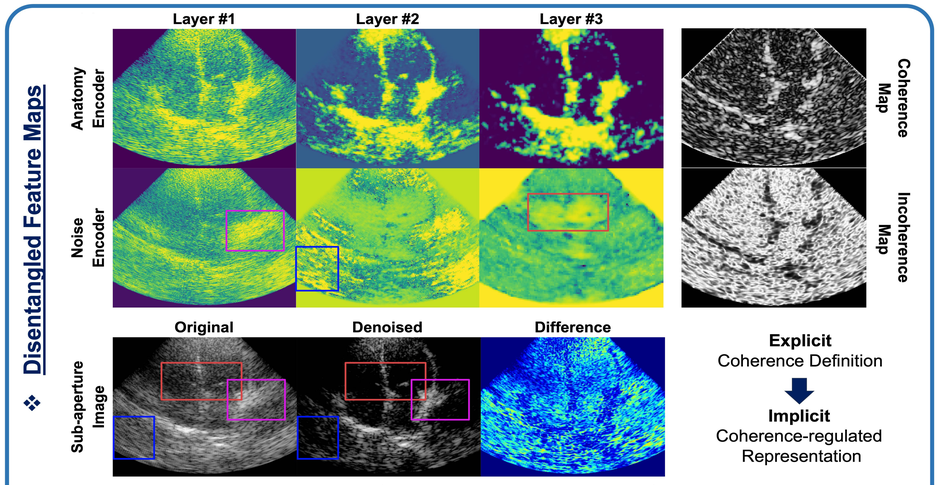


❖ **Line Target Profile**

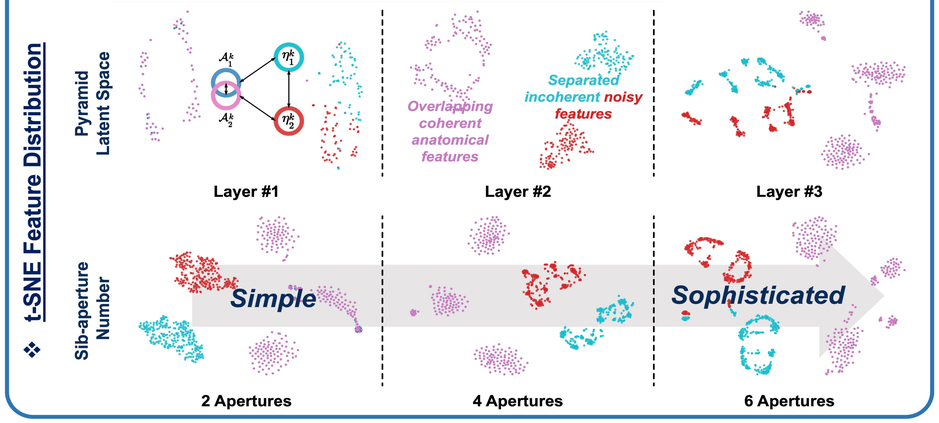


❑ Discussion

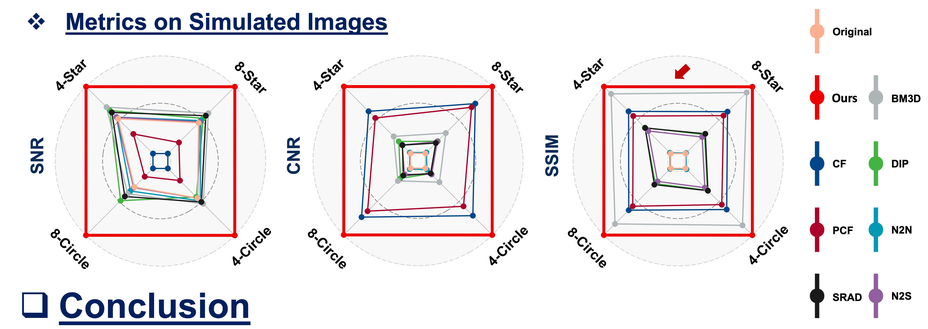
❖ **Disentangled Feature Maps**



❖ **t-SNE Feature Distribution**



❖ **Metrics on Simulated Images**



❑ Conclusion

- ❖ Coherence and incoherence components can be learnt by neural networks → Multi-scale, Robustness, Generalizability
- ❖ Appliable to SAU, multi-angle PW or DW → Multi-scale, Robustness, Generalizability
- ❖ Pre-training free + Pure Test-time Training → Domain shift ✓
- ❖ Self-supervision + Single-shot Noisy Data → Labeled data reliance ✓